

Revisiting Local Context for Long-Horizon Streaming 3D Reconstruction

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Abstract

Streaming 3D reconstruction from extremely long videos requires estimating camera motion and scene geometry online under bounded memory and computation. Early streaming models achieve causal, bounded-cost inference using finite context buffers or compact recurrent states, yet their estimates often deteriorate as sequences grow. Recent methods improve long-horizon stability by coupling short-range context with persistent or multi-level long-range memory. We pursue a different route: we keep the learned temporal state strictly local and formulate predictions whose targets remain independent of sequence length. We present *ABot-Recon*, a simple streaming model that caches KV features from only the preceding 11 frames. It predicts a point map in the current camera coordinate system together with an adjacent-frame relative pose. These predictions remain equivariant under changes of reference frame, and global poses and geometry are recovered through sequential composition. To reduce accumulated drift, a lightweight temporal refiner improves relative rotations using recent visual and motion context, while a composition-aware pose loss supervises multi-step pose composition. Extensive evaluations on challenging long-sequence benchmarks demonstrate the superior long-horizon performance of our local-context approach. On Oxford Spires, *ABot-Recon* achieves an ATE of 4.35 m and an RPE-R of 0.12°, reducing both errors by approximately 40% relative to the best prior results.

Project Page: <https://amap-cvlab.github.io/ABot-Recon-html>

Code: <https://github.com/amap-cvlab/ABot-Recon>

1 Introduction

Recovering camera motion and dense 3D scene geometry from a video stream is a fundamental capability for robotics, autonomous driving, and embodied intelligence. Recent feed-forward reconstruction models [1–4] have shown that point maps and camera parameters can be inferred directly from unposed images using learned geometric priors. Most of these models, however, operate on a fixed collection of views through pairwise or global interactions. Ultra-long video streams change the nature of the problem: observations arrive causally, memory and per-frame computation must remain bounded, and the current view can be separated from early reference frames by thousands of time steps and substantial camera displacement.

Existing streaming reconstruction models adapt learned geometric priors to online inference by propagating historical information through recurrent representations, cached features, or explicit spatial memories [5–15]. To sustain reconstruction over increasingly long sequences, these methods employ various mechanisms to retain, compress, select, and update historical context. More recently, LingBot-Map [16] and HorizonStream [17] have pushed streaming reconstruction beyond 10,000

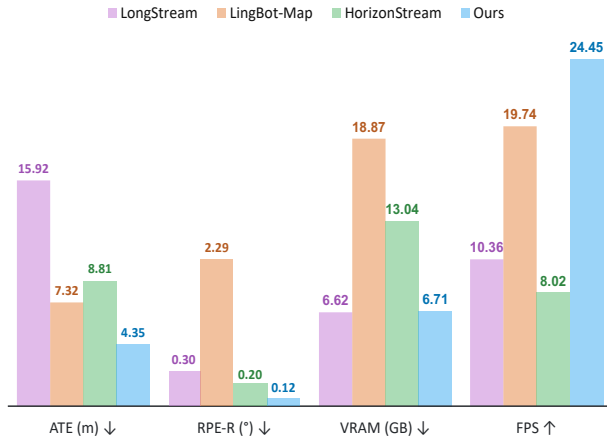


Figure 1. Pose accuracy on Oxford Spires and streaming inference efficiency on KITTI-02, measured by ATE, RPE-R, FPS, and peak GPU memory excluding input storage, on an NVIDIA H100 GPU.



Figure 2. Results with optional loop closure. Our method handles diverse platforms and environments, including large-scale outdoor driving, hand-held indoor traversal, and quadruped robot locomotion.

frames by coordinating short-range observations with persistent long-range information. These advances demonstrate the effectiveness of long-range context, while making the management and fusion of historical state a central part of the architecture.

We focus on a different aspect of the problem: how geometric predictions are parameterized over time. When pose and geometry targets are defined relative to a temporally distant reference, their estimation range grows with the sequence, making the prediction problem increasingly difficult. We instead keep the learned temporal context strictly local and formulate prediction targets in local reference frames over fixed temporal ranges. This formulation brings two benefits: it removes the need for elaborate long-range cache management, and it keeps the per-step prediction problem unchanged as the stream grows. The model can therefore be trained on short sequences and applied to ultra-long streams, while the global trajectory and reconstruction are assembled incrementally from composable local measurements.

Based on this principle, we introduce *ABot-Recon*, a streamlined streaming reconstruction model whose learned components operate within a 12-frame temporal horizon. For each incoming frame, the model attends only to KV features cached from the preceding 11 frames and maintains no persistent learned long-range state. *ABot-Recon* predicts a point map P_i in the current camera coordinate system, a confidence map S_i , and an adjacent-frame transformation $T_{i-1 \leftarrow i}$. Its local predictions transform consistently under changes of reference frame, avoiding dependence on a fixed global anchor. Global camera poses are recovered by chaining the adjacent transformations, and the local point maps are transformed accordingly into the global reconstruction. This design keeps model memory and per-frame computation independent of the total sequence length.

However, the core challenge is error accumulation: small errors in adjacent-frame pose estimates, particularly rotations, can compound into substantial trajectory drift over long sequences. We address this problem at both the estimation and supervision levels. A lightweight temporal rotation refiner combines recent motion evolution with dense visual evidence to improve relative rotation estimates. A composition-aware pose objective supervises transformations over multiple temporal spans, optimizing local predictions according to their accumulated effect rather than treating each adjacent pair in isolation. Together, these components reduce long-horizon drift without extending temporal context.

Extensive experiments show that *ABot-Recon* generalizes from short training sequences to ultra-long video streams and delivers superior camera tracking and 3D reconstruction performance on challenging long-sequence benchmarks, including VBR [18], KITTI [19], Oxford Spires [20], and others. On Oxford Spires, *ABot-Recon* reduces both ATE and RPE-R by approximately 40% relative to the respective best prior results. Since *ABot-Recon* produces a stream of local geometric measurements, it can be readily coupled with standard SLAM modules at inference time without retraining. Adding a loop-closure backend further reduces the ATE to 4.02 m. These results demonstrate that a learned model with only local temporal context can recover accurate and stable trajectories over extended spatial and temporal horizons.

Our contributions are summarized as follows:

- We establish a local-context formulation for ultra-long streaming 3D reconstruction, in which prediction targets remain local and their estimation ranges remain fixed as the sequence grows.
- We introduce *ABot-Recon*, a streamlined streaming model that operates within a 12-frame temporal horizon, consisting of the current frame and KV features from the preceding 11 frames, and recovers global poses and geometry from reference-frame-consistent local predictions.
- We stabilize long-horizon pose composition through a lightweight temporal rotation refiner and composition-aware pose supervision, achieving superior performance on challenging long-sequence benchmarks without persistent learned long-range memory.

2 Related Work

2.1 Feed-Forward Visual Geometry Reconstruction

Classical structure-from-motion (SfM) and visual SLAM systems [21, 22] recover camera motion and scene structure through carefully engineered pipelines that combine correspondence estimation, geometric verification, triangulation, bundle adjustment, and loop closure. Recent feed-forward methods replace a substantial part of this pipeline with learned geometric priors. DUST3R [1] introduced pointmap regression in a shared coordinate frame, enabling two-view reconstruction without known camera intrinsics or poses, while MAST3R [2] augmented this representation with dense local features for accurate image matching.

Subsequent methods extended this paradigm to broader reconstruction settings. MV-DUST3R+ [23] and Fast3R [24] directly exchange information across multiple input images, while VGGT [3] jointly predicts cameras, depth, point maps, and tracks through global multi-view interaction. Pi3 [4] imposes permutation equivariance over the input views, FLARE [25] couples camera estimation with geometry and appearance prediction, PLANA3R [26] represents indoor scenes using metric planar primitives learned through planar splatting, and MonST3R [27] extends feed-forward reconstruction to dynamic videos. These models demonstrate that strong learned priors can replace much of the conventional reconstruction pipeline. However, they either process a bounded collection of views jointly or require pairwise or global alignment, and therefore do not by themselves provide causal inference with bounded cost over an indefinitely growing video stream.

2.2 Streaming Reconstruction and Temporal Memory

Streaming reconstruction propagates information from previous observations through different forms of temporal state. Spann3R [5] stores geometric features in an external spatial memory, CUT3R [6] maintains a compact recurrent state, and SLAM3R [10] registers point maps reconstructed from overlapping local clips. Causal Transformer approaches include StreamVGGT [7] with cached temporal keys and values, and STream3R [8] with decoder-only causal prediction. WinT3R [9] combines sliding-window interaction with global camera tokens, while Point3R [28] uses an explicit spatial pointer memory.

To extend these architectures to longer sequences, subsequent methods strengthen how temporal state is retained and updated. LONG3R [11] introduces gated 3D spatio-temporal memory and length-increasing curriculum training. LongStream [12] adopts keyframe-relative prediction and periodic cache refresh to scale beyond short clips. STAC [13] compresses cached features, TTT3R [14] applies confidence-adaptive recurrent-state updates, and Mem3R [15] decouples tracking and mapping through fast-weight memory and a fixed-size token state.

LingBot-Map [16] and HorizonStream [17] further extend reconstruction beyond 10,000 frames by explicitly coordinating short- and long-range information. LingBot-Map combines anchor context, a pose-reference window, and trajectory memory, whereas HorizonStream couples local attention with persistent linear attention. We instead retain only the KV cache of the most recent K frames, without persistent or hierarchical long-range memory, and formulate outputs as local geometric measurements whose prediction range does not grow with the sequence.

2.3 Local-to-Global Pose and Trajectory Recovery

Separating local motion estimation from global trajectory recovery is central to visual odometry and SLAM. DROID-SLAM [29] estimates dense inter-frame constraints and optimizes them over a graph, while ORB-SLAM3 [22] combines keyframes, bundle adjustment, and loop closure. Learned systems adopt a similar decomposition. MAST3R-SLAM [30] uses pointmap matching for tracking, loop closure, and global optimization, whereas SLAM3R [10], VGGT-Long [31], and TALO [32] construct and align local submaps over time. Anchor3R [33] integrates current-centered relative poses from transient windows through a motion graph. R3 [34] improves pose estimation through relative prediction but remains tied to the first-frame reference, while LoGeR [35] combines chunk-wise inference with test-time global-consistency optimization. In these methods, global consistency is primarily recovered through an additional registration, optimization, or adaptation stage.

We instead incorporate local-to-global decomposition into the learned streaming predictor. The model predicts poses over bounded local prefixes rather than in a coordinate frame tied to a distant first frame, and adopts a reference-frame-equivariant architecture. A pose-chain loss supervises compositions across multiple temporal spans, while rotation refinement uses only recent visual and motion context. Both components preserve bounded memory while improving long-horizon composition. An optional, training-independent loop-closure backend can further refine the trajectory at inference time. Thus, long-horizon capability arises from local prediction and reliable composition rather than learned access to long-range memory.

3 Method

3.1 Overview

Our key design principle is to formulate long-horizon streaming reconstruction as local geometric estimation, thereby enabling a simple and scalable architecture, as shown in Figure 3. For each incoming frame I_i , *ABot-Recon* predicts a point map P_i in the current camera coordinate system, a corresponding confidence map S_i , and a relative pose $T_{i-1 \leftarrow i}$ to the previous frame. The model operates with a fixed K -frame temporal context, set to 12 frames in our implementation, and maintains no persistent learned long-range state. Global camera poses and scene geometry are recovered by composing these local predictions over time, keeping model memory and per-frame computation independent of sequence length.

Section 3.2 describes the local prediction formulation and windowed causal attention. Section 3.3 introduces a lightweight temporal refiner that improves relative rotations using recent motion and visual context. Section 3.4 presents a composition-aware objective that supervises pose chains over multiple temporal spans.

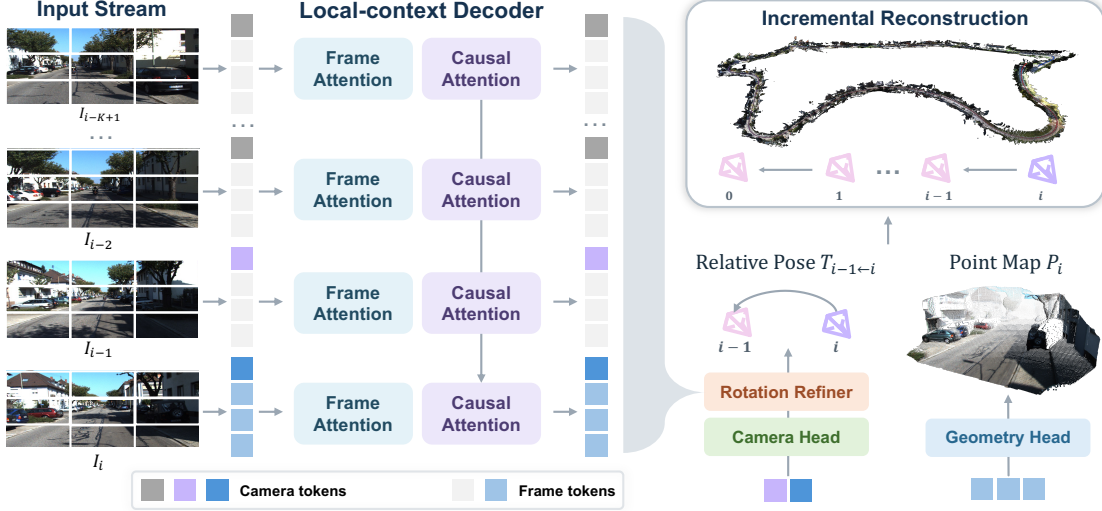


Figure 3. Overview of our method. For each incoming frame I_i , our model makes predictions using a fixed K -frame causal context. A shared image encoder followed by alternating intra-frame and windowed causal cross-frame attention predicts dense frame tokens and compact camera tokens. The geometry head predicts a local point map as well as the corresponding confidence map in the current camera coordinate system, while adjacent camera-token descriptors are used to predict the relative pose. A lightweight motion-visual rotation refiner combines the motion and visual evidence to estimate a residual rotation and the refined adjacent transformations are composed online to recover the global camera trajectory.

3.2 Local Geometry Estimation

For each incoming frame I_i , our model predicts a point map P_i in the current-frame coordinate system, a corresponding confidence map S_i , and the relative pose $T_{i-1 \leftarrow i}$ of the current frame I_i with respect to the previous frame I_{i-1} . Our model builds on a causal-attention Transformer backbone, formulated as

$$\begin{aligned} F_i &= \text{Encoder}(I_i), \\ (G_i, C_i, \mathcal{M}_i) &= \text{Decoder}([F_i, C], \mathcal{M}_{i-1}), \\ P_i &= \text{Head}_{\text{point}}(G_i), \\ S_i &= \text{Head}_{\text{conf}}(G_i), \end{aligned} \quad (1)$$

where C denotes a set of learnable camera tokens, $C_i = \{c_i^\ell\}_{\ell=1}^L$ denotes their decoded representations for frame I_i and $\mathcal{M}_{i-1}$ denotes the memory caches.

The relative pose is predicted directly from the camera tokens of two adjacent frames. We first aggregate the camera tokens of the same frame into a frame-level pose descriptor by an MLP ϕ_{desc} ,

$$z_i = \frac{1}{L} \sum_{\ell=1}^L \phi_{\text{desc}}(c_i^\ell), \quad (2)$$

and construct a pairwise descriptor [36] between adjacent frames

$$q_i = \mathcal{R}(z_{i-1}, z_i), \quad \mathcal{R}(x, y) = [x, y, y - x, x \odot y]. \quad (3)$$

The adjacent-frame relative pose is then obtained as

$$T_{i-1 \leftarrow i} = \text{Head}_{\text{pose}}(q_i). \quad (4)$$

The relative pose between any two frames I_i and I_j with $i < j$ can be recovered by chaining the predicted adjacent-frame transformations:

$$T_{i \leftarrow j} = T_{i \leftarrow i+1} T_{i+1 \leftarrow i+2} \cdots T_{j-1 \leftarrow j} = \prod_{k=i+1}^j T_{k-1 \leftarrow k}. \quad (5)$$

Since geometric prediction is performed locally, we can further restrict the temporal context to a fixed-size window. Specifically, we replace full-history causal temporal attention with windowed

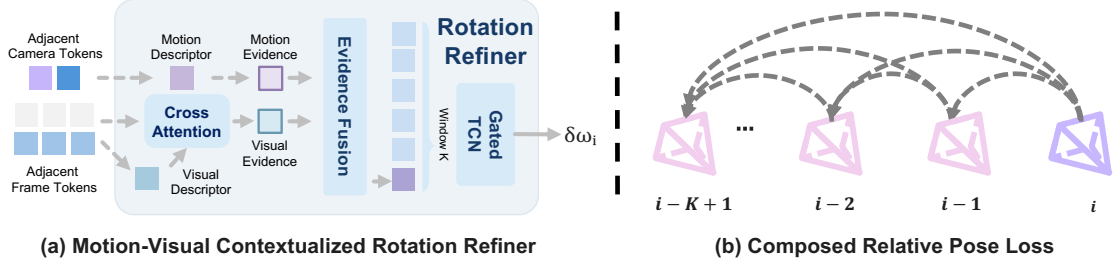


Figure 4. Motion-visual contextualized rotation refiner with composition-aware pose supervision. **(a)** For each pair of adjacent frames, motion evidence is encoded from the camera tokens, while a visual descriptor queries the corresponding dense frame tokens to extract visual evidence. The two evidence streams are fused into a unified pairwise representation. These pairwise representations are then aggregated within a causal temporal window by a lightweight gated TCN, which predicts a rotation residual to refine the initial relative rotation. **(b)** Rather than supervising adjacent transformations independently, we compose the refined relative poses over multiple temporal gaps and compare each composed transformation with its ground-truth counterpart. This composition-aware supervision directly exposes accumulated pose-chain errors during training.

causal temporal attention, where frame I_i attends only to the cached keys and values of the most recent $K - 1$ frames:

$$\mathcal{M}_{j-1}^{(K)} = \{\text{KV}_i\}_{i=\max(0, j-K+1)}^{j-1}. \quad (6)$$

Cached states are discarded once they fall outside the window, resulting in $O(K)$ temporal memory and $O(NK)$ temporal-attention computation over a sequence of N frames. For a fixed window size K , inference therefore requires constant memory while computation grows linearly with the sequence length. As this design only restricts the attention context, it remains easy to train and fully compatible with standard attention implementations, yielding a simple and scalable streaming backbone.

3.3 Rotation Refiner

Although predicting relative poses between adjacent frames avoids direct extrapolation in the global coordinate system, it can still be unreliable under rapid camera motion or challenging visual conditions. Moreover, it fails to enforce temporal consistency in the motion estimates, allowing errors to accumulate and causing long-term drift during recursive pose composition. To mitigate this issue, we introduce a Motion-Visual Contextualized Rotation Refiner that leverages recent motion history to refine unreliable relative pose estimates before trajectory composition. Since adjacent-frame translation is generally more constrained and stable than rotation, we preserve the predicted translation and apply temporal refinement only to the rotational component.

As shown in Figure 4(a), for each pair of adjacent frames, we construct a unified motion-visual evidence representation by integrating compact inter-frame motion evidence with spatially resolved visual evidence. Specifically, the motion evidence f_m is obtained by encoding the predicted pairwise motion descriptor q_i using an MLP ϕ_m :

$$f_m = \phi_m(q_i). \quad (7)$$

Here, ϕ_m , ϕ_v , ϕ_{fuse} , and ϕ_o denote the MLPs used for motion encoding, visual-query encoding, motion-visual fusion, and rotation-residual prediction, respectively.

To complement the global motion evidence, we extract pair-specific visual evidence in a coarse-to-fine manner. We first average-pool the dense patch tokens of each frame to obtain $\bar{G}_{i-1}$ and $\bar{G}_i$, which are then used to construct a compact visual descriptor. The resulting coarse descriptor is used as a query to retrieve fine-grained information from the original dense frame tokens:

$$f_v = \text{CrossAttn}(\phi_v(\mathcal{R}(\bar{G}_{i-1}, \bar{G}_i)), [G_{i-1}; G_i]), \quad (8)$$

where $\mathcal{R}(\cdot, \cdot)$ denotes the operation of pairwise descriptor. The resulting relational descriptor provides a coarse characterization of the visual variation between the two frames, while cross-attention [37] retrieves the corresponding spatially resolved information from the dense features, yielding the pair-conditioned visual evidence f_v .

Finally, the motion and visual features are fused to yield a compact pairwise representation f_i that jointly captures the predicted camera motion and its supporting visual context:

$$f_i = \phi_{\text{fuse}}([f_m; f_v]). \quad (9)$$

We then jointly model the current pairwise feature and its $K - 1$ immediately preceding features, forming a causal temporal window of length K . A lightweight gated temporal convolutional network (TCN) [38] is used to aggregate the recent motion-visual context:

$$\begin{aligned} \mathcal{W}_i &= [f_{i-K+1}, \dots, f_i], \\ \delta\omega_i &= \phi_o(\mathcal{T}_h(\mathcal{W}_i) \odot \sigma(\mathcal{T}_g(\mathcal{W}_i))). \end{aligned} \quad (10)$$

Here, $\mathcal{T}_h$ and $\mathcal{T}_g$ denote two causal TCN branches responsible for temporal feature aggregation and adaptive gating, respectively. By contextualizing the current pairwise estimate with recent motion-visual dynamics, the refiner predicts a residual that compensates for unreliable local rotation estimates while preserving causality for online inference.

The resulting axis-angle residual $\delta\omega_i \in \mathbb{R}^3$ is mapped to $\text{SO}(3)$ via the exponential map and composed with the initial relative rotation:

$$\hat{R}_{i-1 \leftarrow i} = \tilde{R}_{i-1 \leftarrow i} \text{Exp}([\delta\omega_i]_{\times}). \quad (11)$$

Guided by the temporal motion-visual context, this residual update corrects unreliable relative-rotation estimates. The refined relative poses are then composed to recover the camera trajectory.

3.4 Composition-aware Loss Functions

To promote consistency under recursive pose composition and mitigate error accumulation, we introduce multi-gap relative-pose supervision. Rather than constraining only adjacent-frame predictions, it supervises transformations composed across multiple temporal gaps. Specifically, for a sequence of N frames, we define the set of supervised frame pairs as

$$\mathcal{P} = \{(i, j) \mid 0 \leq i < j < N, j - i \leq K - 1\}, \quad (12)$$

where K denotes the window size. For each $(i, j) \in \mathcal{P}$, we supervise the composed transformation against the corresponding ground-truth transformation, as illustrated in Figure 4(b). To emphasize errors accumulated over longer composition chains, we assign each frame pair a weight proportional to its temporal gap and normalize the weights over all supervised pairs:

$$\alpha_{ij} = \frac{(j - i)^\gamma}{\frac{1}{|\mathcal{P}|} \sum_{(m, n) \in \mathcal{P}} (n - m)^\gamma}, \quad 0 < \gamma < 1. \quad (13)$$

Following π^3 [4], we use the same translation and rotation losses for each composed relative pose:

$$\mathcal{L}_{\text{pose}} = \frac{1}{|\mathcal{P}|} \sum_{(i, j) \in \mathcal{P}} \left(\lambda_{\text{trans}} \ell_{\text{trans}}^{(i, j)} + \alpha_{ij} \lambda_{\text{rot}} \ell_{\text{rot}}^{(i, j)} \right). \quad (14)$$

We additionally introduce a residual regularization term, $\mathcal{L}_{\text{smooth}}$, that penalizes the magnitude and temporal variation of the predicted rotation residuals, thereby encouraging small-magnitude and temporally consistent refinements.

For local geometric supervision, we retain the point-map loss $\mathcal{L}_{\text{pts}}$, surface-normal loss $\mathcal{L}_{\text{normal}}$, and confidence loss $\mathcal{L}_{\text{conf}}$ [4], which supervise 3D structure, surface orientation, and pointwise prediction reliability, respectively.

The overall training objective is

$$\mathcal{L} = \lambda_{\text{pose}} \mathcal{L}_{\text{pose}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}} + \lambda_{\text{pts}} \mathcal{L}_{\text{pts}} + \lambda_{\text{normal}} \mathcal{L}_{\text{normal}} + \lambda_{\text{conf}} \mathcal{L}_{\text{conf}}, \quad (15)$$

where the λ coefficients balance the corresponding supervision terms.

4 Experiments

4.1 Experimental Setup

Training data. We train on a diverse mixture of 30 synthetic and real-world datasets covering indoor scenes, outdoor environments, autonomous driving, handheld capture, and aerial trajectories. As summarized in Table 1, synthetic and real-world data account for 62.05% and 37.95% of the sampling distribution, respectively. We assign larger sampling probabilities to datasets containing long, temporally coherent trajectories, while sampling from the remaining datasets to preserve diversity in scene appearance, geometry, and camera motion.

Table 1. Training-data composition. Sampling ratios are obtained by normalizing the dataset weights used by the training sampler. Internal data are divided into two categories: *Internal Synthetic* and *Internal Real*.

Synthetic data		Real-world data	
Dataset	Ratio (%)	Dataset	Ratio (%)
TartanAir-v2 [39]	8.96	DL3DV [40]	8.96
TartanAir [39]	6.72	Waymo [41]	5.38
OmniWorld-Game [42]	8.96	ARKitScenes [43]	1.34
TartanGround [44]	6.72	ScanNet++ [45]	3.81
VirtualKITTI2 [46]	6.27	WildRGBD [47]	2.91
HyperSim [48]	6.27	ScanNet [49]	2.46
PointOdyssey [50]	0.90	MapFree [51]	1.12
Unreal4K [52]	1.34	ARKitScenes HR [53]	2.46
Spring [54]	0.23	UASOL [55]	0.45
MVS-Synth [56]	0.67	DDAD [57]	1.34
Dynamic Replica [58]	0.45	KITTI-360 [59]	2.02
ASE [60]	2.69	Holo360D [61]	2.02
MidAir [62]	2.69	Internal Real	3.68
MatrixCity [63]	1.34		
AirZoo [64]	1.34		
BlendedMVS [65]	2.47		
Internal Synthetic	4.03		
Synthetic total	62.05	Real-world total	37.95

Table 2. Evaluation datasets and statistics.

Dataset	Task	Frames/sequence	#Seq.	Inference stride
KITTI [19]	Camera pose	271–4,661	11	1
VBR [18]	Camera pose	8,815–18,846	7	1
Oxford Spires [20]	Pose / Reconstruction	3,821–3,840	10	1
7Scenes [66]	Reconstruction	500–1,000	7	1
TUM-Dynamic [67]	Reconstruction	707–1,261	8	1

Data processing and augmentation. We convert all camera poses to a unified camera-to-world convention, normalize dataset-specific depth and translation units, and discard samples with invalid poses, missing frames, or corrupted geometry. For video datasets, frames are sampled in their original temporal order using dataset-specific intervals. For unordered multiview datasets, we construct pseudo-sequences from a camera pose graph that favors locally coherent neighboring views. We apply focal-length and crop perturbations and preserve the source field of view by resizing to the target width followed by center cropping or mean-color padding. For datasets with notably off-center cameras, we retain the original non-central principal point with probability 0.9 instead of always applying principal-point-centered cropping. Photometric augmentation includes brightness, contrast, saturation, hue, and gamma jitter, as well as JPEG degradation and image blur. Dataset-specific temporal intervals and augmentation parameters are provided in the supplementary material.

Training stages. Training consists of two main stages with a progressively extended temporal horizon, followed by a short confidence-calibration phase. We initialize Stage I from the publicly released π^3 weight [4]. We replace the original absolute camera-pose output with our adjacent-frame relative pose parameterization and adapt the model to causal streaming inference, without introducing the confidence branch. Stage I uses 32-frame clips and is trained for 32K iterations with a batch size of 48 on 48 NVIDIA H20 GPUs. This stage establishes stable local geometry and adjacent motion estimation.

In Stage II, we initialize from the Stage I checkpoint and extend the training clips from 32 to 128 frames, while keeping the local prediction window fixed at 12 frames. Longer clips provide denser temporally continuous pose supervision and expose the model to more complete pose-composition chains within each training sample. We also enable the motion-visual contextualized rotation refiner module, which refines local rotation estimates before they are composed into the trajectory. Stage II is trained for another 38K iterations with a batch size of 32 on 32 AMD MI308 GPUs.

Finally, we introduce the confidence prediction branch and fine-tune it for an additional 4K iterations. During this phase, the reconstruction and pose estimation networks are kept fixed, and only the confidence branch is optimized. This confidence-only fine-tuning allows the model to estimate the reliability of its dense predictions without altering the geometry and camera trajectories learned in the preceding stages.

Implementation details. We use the AdamW optimizer [68] throughout training. In Stage I, the learning rate reaches a peak of 1×10^{-4} and is gradually annealed to 1×10^{-6} . In Stage II, we

Table 3. Camera pose estimation results on KITTI. Lower is better for all metrics. Within streaming methods, best and second-best results are marked in blue and underlined, respectively. * denotes methods that require camera intrinsics, and ‡ denotes results quoted from the corresponding paper rather than reproduced under our evaluation pipeline. w/ LC indicates results with loop closure enabled.

		Per-sequence ATE (m) ↓										Sequence averages			
		00	01	02	03	04	05	06	07	08	09	10	Avg. ATE (m)↓	Avg. RPE _r (°)↓	Avg. RPE _t (m)↓
Group	Method	4541f	1101f	4661f	801f	271f	2761f	1101f	1101f	4071f	1591f	1201f			
		3.70 km	2.50 km	5.10 km	0.60 km	0.40 km	2.20 km	1.20 km	0.70 km	3.20 km	1.70 km	0.90 km			
Opt.-based	VGGT-SLAM	27.99	214.23	260.88	53.50	26.02	99.61	23.62	64.52	243.87	169.10	80.13	114.86	1.57	2.73
	MASt3R-SLAM	190.12	507.15	232.17	169.82	88.98	126.37	133.33	62.92	199.70	178.03	151.69	185.48	0.98	2.58
	VGGT-Long	8.63	58.37	52.79	8.83	4.23	9.83	4.65	2.67	73.12	31.84	27.67	25.69	0.26	0.42
	SLAMFormer-∞ [‡]	15.39	96.58	28.81	4.97	4.16	11.36	13.54	8.53	37.00	18.85	13.93	23.01	–	–
	Droid-SLAM*	131.25	267.79	217.63	3.40	1.47	80.41	64.20	18.19	167.80	90.05	22.74	96.81	0.23	0.60
	DPV-SLAM*	113.19	16.76	113.06	2.44	0.99	59.42	53.32	18.90	110.36	74.60	13.74	52.43	0.05	0.27
	DPVO*	113.17	16.56	113.37	2.45	0.98	59.59	55.63	19.40	110.64	74.44	13.80	52.73	0.05	0.27
	Droid-W*	48.26	207.57	52.80	1.76	4.65	25.16	5.34	3.35	29.05	35.56	8.01	38.32	0.05	0.36
Streaming	InfiniteVGGT	186.48	624.69	289.03	168.42	73.88	143.81	116.65	85.53	225.14	214.28	156.87	207.71	9.44	19.21
	OVGGT	185.65	716.66	304.04	155.78	65.01	155.48	108.51	85.92	246.97	215.64	178.67	219.85	11.02	18.02
	Stream3R-w	188.66	695.02	301.28	161.68	98.99	159.85	122.82	87.19	264.61	219.90	194.32	226.76	7.03	11.52
	CUT3R	185.21	654.37	300.51	157.95	23.72	153.77	133.09	68.57	230.95	206.95	184.34	209.04	2.94	2.79
	CUT3R w/ reset	190.70	91.56	269.58	20.95	5.97	104.88	72.01	27.44	164.69	93.11	41.53	98.40	0.55	0.65
	TTT3R	165.12	524.16	272.38	105.06	12.32	146.24	131.02	61.75	262.88	187.14	124.13	181.11	3.85	3.49
	TTT3R w/ reset	118.19	102.98	242.05	14.37	3.52	31.62	37.66	11.99	81.52	79.10	29.81	68.44	0.49	0.60
	LongStream	90.02	63.46	236.42	5.04	2.57	80.62	12.76	13.84	89.61	95.07	25.67	65.01	0.35	0.33
	LingBot-Map	33.57	151.48	66.02	2.72	2.09	12.79	6.06	4.82	13.36	19.50	8.01	29.13	0.61	3.04
	HorizonStream	29.67	24.15	97.24	5.53	0.72	13.94	5.46	6.12	22.63	29.79	12.35	22.51	0.17	0.11
	HorizonStream w/ LC	15.45	24.15	71.01	5.53	0.72	7.84	7.54	3.14	22.63	27.45	12.35	17.98	0.30	0.09
	Ours	21.86	21.55	69.87	4.38	2.93	13.54	12.04	2.02	22.30	14.11	16.17	18.25	0.10	0.10
Ours w/ LC	11.70	21.55	39.00	4.38	2.93	6.00	8.70	1.65	22.30	13.99	16.17	13.49	0.10	0.10	

adopt differential learning rates: the newly introduced rotation refiner module is optimized with a peak learning rate of 5×10^{-5} , while the remaining model parameters are fine-tuned with a smaller peak learning rate of 2×10^{-5} . This setting allows the new module to adapt effectively while limiting disruption to the local pose and geometry representations learned in Stage I. During the final confidence-calibration phase, only the confidence branch is optimized, using a peak learning rate of 2×10^{-5} .

We maintain an exponential moving average (EMA) [69] with a decay coefficient of 0.999 throughout training. Parameter averaging smooths the optimization trajectory and reduces sensitivity to transient noisy or unusually large updates, which is particularly useful for our heterogeneous training mixture. We use a per-GPU batch size of one sequence, bfloat16 mixed precision, and gradient clipping at 1.0.

Input images are resized to a width of 504 pixels while preserving their aspect ratio. The resulting images are then center-cropped or mean-padded vertically to obtain an input resolution of 504×280 . The same causal inference rule is applied to all test sequences.

Inference mode. At inference time, the model processes the input stream causally and continuously, without chunk-wise resets. Since its learned streaming state consists only of a fixed-size local KV cache, inference does not require the total number of frames to be known in advance, and the same procedure applies regardless of sequence length. The causal KV cache is implemented using the paged KV-cache operators in FlashInfer [70] for efficient streaming attention. Each incoming frame is processed once to predict its local geometry and adjacent-frame relative pose, after which the global camera trajectory and point cloud are recovered incrementally through pose composition.

ABot-Recon predicts relative poses from local frame windows, allowing the same prediction interface to provide both sequential odometry constraints and relative-pose constraints between revisited locations. The base model operates without additional long-range memory or online optimization. When stronger global consistency is required, we optionally attach a training-independent SLAM-style loop-closure backend [17, 31], which refines the accumulated trajectory without altering the underlying causal prediction process.

To detect loop closures, revisited frame pairs are retrieved from historical observations using FAISS-based search [71] over DINOv2-SALAD descriptors [72]. For each retrieved pair, local windows centered on the two frames are jointly fed into *ABot-Recon* to estimate their relative pose. The sequential predictions and retrieved-pair predictions are respectively converted into odometry and loop-closure constraints. These constraints are incorporated into a sparse pose graph, whose optimization refines the global trajectory.

Evaluation datasets. We evaluate our method on two complementary tasks: camera pose estimation and dense 3D reconstruction. For camera pose estimation, we use KITTI [19], Oxford Spires [20],

Table 4. Camera pose estimation on Oxford Spires. Within streaming methods, best and second-best results are marked in blue and underlined, respectively. * denotes methods that require camera intrinsics.

Group	Method	Per-sequence ATE (m) ↓										Sequence averages		
		Keb1e-2	Keb1e-3	Keb1e-4	Keb1e-5	Observ.-1	Observ.-2	Christ-2	Christ-3	Christ-5	Bodlelan-2	Avg.	Avg.	Avg.
		3840f 0.20 km	3840f 0.18 km	3840f 0.23 km	3840f 0.21 km	3840f 0.27 km	3839f 0.28 km	3821f 0.25 km	3840f 0.18 km	3840f 0.80 km	3840f 0.55 km	ATE (m) ↓	RPE _r (°) ↓	RPE _t (m) ↓
Opt.-based	VGGT-SLAM	2.64	3.78	3.89	1.86	5.74	3.39	7.41	1.02	12.94	18.83	6.15	0.39	0.05
	MASt3R-SLAM	7.06	18.03	11.27	14.64	10.26	10.30	2.90	6.95	19.85	19.99	12.12	0.36	0.12
	VGGT-Long	9.07	9.37	11.77	3.96	6.03	4.05	2.41	0.48	14.38	12.59	7.41	0.29	0.08
	Droid-SLAM*	16.79	15.99	29.40	21.00	0.87	1.08	1.31	1.15	7.58	1.82	9.70	0.12	0.03
	DPV-SLAM*	0.20	1.31	0.03	0.30	0.16	0.45	0.15	0.04	25.77	1.08	2.95	0.15	0.03
	DPVO*	0.12	0.84	0.05	0.29	0.16	0.51	0.15	0.35	12.04	1.08	1.56	0.16	0.01
Streaming	Droid-W*	2.45	0.99	2.64	4.27	1.07	1.06	1.63	0.75	1.56	6.28	2.27	0.08	0.01
	InfiniteVGGT	25.02	20.39	25.09	24.15	29.10	28.14	33.02	12.95	40.39	79.32	31.76	5.30	2.70
	OVGGT	32.97	23.01	28.53	23.23	28.51	27.75	30.01	13.09	42.35	82.41	33.19	7.83	3.61
	Stream3R-w	31.28	22.39	31.07	24.63	28.38	27.18	38.12	12.99	42.39	82.66	34.11	9.05	2.67
	CUT3R	31.68	21.67	22.76	25.07	26.77	29.41	33.93	13.06	43.66	79.93	32.79	1.78	0.25
	CUT3R w/ reset	13.10	11.55	15.76	8.98	13.58	15.60	12.36	5.71	31.31	19.42	14.74	0.66	0.05
	TTT3R	24.66	22.56	27.33	18.67	26.94	28.18	19.83	9.68	26.76	55.01	25.96	3.19	0.31
	TTT3R w/ reset	8.27	7.26	8.77	7.55	9.43	10.30	4.56	2.11	<u>21.75</u>	14.00	9.40	0.29	0.04
	LongStream	11.57	9.44	16.88	10.67	12.43	14.34	17.93	6.42	35.49	24.02	15.92	0.30	0.06
	LingBot-Map	2.71	3.04	2.04	3.23	5.38	4.43	2.21	0.74	37.99	11.43	7.32	2.29	0.53
	HorizonStream	5.79	6.47	7.36	6.27	6.33	5.30	10.69	3.11	25.35	<u>11.41</u>	8.81	<u>0.20</u>	<u>0.03</u>
	HorizonStream w/ LC	5.79	6.47	1.25	6.24	6.33	5.30	10.69	0.47	25.35	<u>11.41</u>	7.93	<u>0.20</u>	<u>0.03</u>
	Ours	<u>4.24</u>	1.70	5.08	<u>4.44</u>	4.60	4.40	<u>3.07</u>	1.15	9.03	5.76	<u>4.35</u>	0.12	0.02
	Ours w/ LC	<u>4.24</u>	1.70	<u>2.04</u>	<u>4.44</u>	4.60	4.40	<u>3.07</u>	0.85	9.03	5.76	<u>4.02</u>	0.12	0.02

and VBR [18]. These benchmarks cover kilometer-scale driving sequences as well as handheld and vehicle-mounted capture. For dense 3D reconstruction, we report results on 7Scenes [66], TUM-Dynamic [67], and Oxford Spires [20], covering indoor and outdoor environments as well as static and dynamic scenes. All methods process every evaluated sequence once in temporal order with a stride of 1. For 7Scenes, we select **seq-01** from each of its seven scenes; for all other benchmarks, we use all sequences in the corresponding evaluation set. Dataset statistics are summarized in Table 2.

Baseline methods. We compare against two groups of methods. Streaming reconstruction baselines include LingBot-Map [16], HorizonStream [17], LongStream [12], InfiniteVGGT [73], OVGGT [74], CUT3R [6], TTT3R [14], and Stream3R-w [8]. We additionally compare camera trajectories with optimization-based methods including VGGT-SLAM [75], MASt3R-SLAM [30], VGGT-Long [31], SLAMFormer- ∞ [76], Droid-SLAM [29], DPV-SLAM [77], DPVO [78], and Droid-W [79]. Droid-SLAM, DPV-SLAM, DPVO, and Droid-W use camera intrinsics, whereas our method estimates camera motion directly from the RGB stream without requiring test-time intrinsics. Since SLAMFormer- ∞ is not publicly available, we quote its KITTI ATE directly from the paper and all other baselines are evaluated using their publicly released official implementations and checkpoints under our unified protocol. CUT3R and TTT3R process each complete sequence with and without state reset, and Stream3R-w uses its official window mode with a window size of 5.

Evaluation protocol and metrics. All methods process each test sequence in temporal order and all benchmarks are evaluated with a temporal stride of 1. For camera pose evaluation, each predicted trajectory is aligned to the ground truth using Umeyama Sim(3) alignment. We report the RMSE of Absolute Trajectory Error (ATE), relative translational pose error (RPE_t), and relative rotational pose error (RPE_r). For dense reconstruction, predicted point maps are first resampled to the ground-truth evaluation grid using nearest-neighbor interpolation. Following LingBot-Map [16], the predicted and ground-truth point clouds are aligned using Umeyama registration, voxelized, and refined with point-to-point ICP. We use voxel sizes of 4/512 m for 7Scenes and TUM-Dynamic and 0.05 m for Oxford Spires, and report Chamfer Distance (CD) and F1 at a distance threshold of 0.25 m (7Scenes and TUM-Dynamic) and 4 m (Oxford Spires). For Oxford Spires, inference is performed on every frame, while point-cloud evaluation uses every 10th frame to keep evaluation tractable. Lower is better for all metrics except F1.

4.2 Camera Pose Estimation

Overall results. Tables 3, 4, and 5 report camera pose estimation results on KITTI, Oxford Spires, and VBR, respectively. Across the three benchmarks, our method achieves strong long-horizon trajectory accuracy using only a compact local temporal context, without maintaining a persistent global reference or long-range temporal memory.

On KITTI, our method achieves the lowest average ATE among the compared methods without

Table 5. Camera pose estimation on VBR. Within streaming methods, best and second-best results are marked in blue and underlined, respectively. * denotes methods that require camera intrinsics.

Group	Method	Per-sequence ATE (m) ↓							Sequence averages		
		Colosseo-0	Campus-0	Campus-1	Pincio-0	Spagna-0	Diag-0	Ciampino-1	Avg.	Avg.	Avg.
		8815f	12042f	11671f	11142f	14141f	10021f	18846f	ATE (m)↓	RPE _r (°)↓	RPE _t (m)↓
		<u>1.45 km</u>	<u>2.73 km</u>	<u>2.95 km</u>	<u>1.27 km</u>	<u>1.56 km</u>	<u>1.02 km</u>	<u>5.20 km</u>			
Opt.-based	VGGT-SLAM	54.30	122.07	117.15	32.48	24.86	33.52	188.30	81.81	1.08	0.31
	MASt3R-SLAM	88.75	116.21	114.37	67.11	54.62	35.67	171.76	92.64	1.25	0.30
	VGGT-Long	44.03	132.67	115.62	64.29	59.04	33.66	179.43	89.82	0.82	0.37
	Droid-SLAM*	98.12	132.85	119.53	63.21	64.20	35.86	198.21	101.71	0.82	0.23
	DPV-SLAM*	86.34	107.85	70.29	51.56	55.60	33.87	105.91	73.06	0.70	0.16
	DPVO*	90.42	109.72	58.49	52.83	57.84	33.84	102.60	72.25	1.34	0.15
	Droid-W*	50.27	125.02	96.89	21.96	22.82	24.59	171.52	73.29	0.58	0.13
Streaming	InfiniteVGGT	89.89	136.76	116.63	77.76	63.35	33.84	202.09	102.90	8.27	4.84
	OVGGT	96.94	137.74	117.39	77.76	62.49	34.84	201.43	104.08	11.43	6.07
	Stream3R-w	92.51	138.38	116.38	77.43	64.57	35.64	202.14	103.87	7.81	2.72
	CUT3R	99.21	136.32	117.92	75.05	62.29	33.66	202.10	103.79	1.50	0.37
	CUT3R w/ reset	76.68	59.51	67.21	48.57	48.55	26.66	169.18	70.91	0.75	0.11
	TTT3R	91.11	129.47	116.59	74.31	60.96	34.39	194.75	100.23	3.58	0.50
	TTT3R w/ reset	73.32	56.51	52.76	37.39	35.33	34.97	175.98	66.61	0.81	0.10
	LongStream	83.01	120.29	106.92	71.39	60.81	30.61	177.51	92.93	0.71	0.13
	LingBot-Map	<u>11.11</u>	21.45	12.39	46.30	35.40	16.16	63.36	29.45	4.42	2.70
	HorizonStream	40.89	27.84	24.28	26.77	28.17	25.39	29.83	29.02	0.60	0.04
	HorizonStream w/ LC	24.66	27.22	<u>8.72</u>	<u>16.29</u>	<u>21.33</u>	7.36	<u>17.92</u>	<u>17.64</u>	<u>0.61</u>	0.04
	Ours	48.12	<u>11.06</u>	17.69	21.01	29.24	28.06	55.82	30.14	0.60	0.04
	Ours w/ LC	<u>16.70</u>	8.10	4.30	5.16	17.96	<u>9.71</u>	8.01	9.99	0.60	0.04

and with loop closure, respectively, while also obtaining strong relative-pose accuracy. On Oxford Spires, our method achieves an average ATE of 4.35 m without loop closure, together with the best RPE_r and RPE_t among the streaming methods. The optional loop-closure backend further reduces the average ATE to 4.02 m. Although some optimization-based methods (DPV-SLAM, DPVO, and Droid-W) obtain lower ATE on this benchmark, these methods require camera intrinsics, whereas ours operates solely on the RGB stream.

On VBR, which consists of substantially longer sequences under diverse capture conditions, our method remains competitive with HorizonStream and LingBot-Map in global trajectory accuracy without loop closure, while achieving relative-pose accuracy comparable to HorizonStream. With the optional loop-closure backend, our method further improves global trajectory accuracy, achieving the lowest average ATE among the compared methods.

Qualitative trajectory comparison. Figure 5 provides a complementary view of trajectory quality beyond aggregate ATE. Across the three benchmarks, our method preserves the overall trajectory geometry while producing smooth local motion. On particularly long trajectories, our relative-pose formulation exhibits less local trajectory jitter than the anchor-based LingBot-Map, while achieving a level of local stability comparable to HorizonStream despite using only a local context.

Long-horizon stability. To quantitatively characterize the local trajectory stability observed above, we first examine how adjacent-frame relative-pose errors evolve as the processing horizon increases. Figure 6 (a) reports the running RMSE of RPE_r and RPE_t on the 4,661-frame KITTI-02 sequence, computed over all adjacent frame pairs observed up to each processed frame. Our method maintains the lowest running error throughout the sequence, with no progressive degradation as more frames are processed. This behavior is consistent with the length-independent nature of our relative-pose formulation: the regression target remains local regardless of the total stream length.

Stable adjacent-frame predictions, however, do not necessarily guarantee accurate long-term pose composition, as local errors can accumulate through repeated composition. We therefore evaluate multi-gap RPE over all KITTI sequences in Figure 6 (b). Our method achieves the lowest average rotational and translation errors at every evaluated gap. The slower error growth at long gaps is consistent with the role of composed-pose supervision, which directly constrains the multi-step predictions used to assemble the streaming trajectory.

Runtime and memory efficiency. We benchmark forward throughput and peak GPU memory, excluding input storage, on a single NVIDIA H100 GPU, as reported in Table 6. The chunk-based modes of LongStream* and HorizonStream* are reported for reference and are not directly comparable to frame-by-frame streaming results. Our method processes the stream frame by frame at 24.45 FPS using 6.71 GB of GPU memory. Under the same frame-by-frame setting, it is faster than LongStream, LingBot-Map, and HorizonStream, while using substantially less memory than

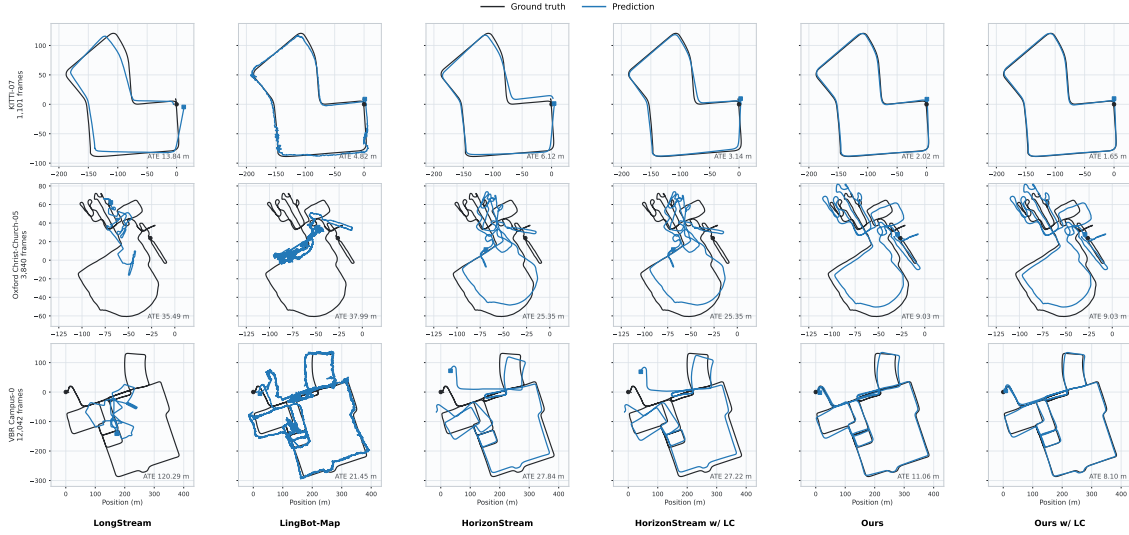


Figure 5. Qualitative camera pose comparison on representative sequences from KITTI, Oxford Spires, and VBR. Predicted trajectories (blue) are aligned to the ground truth (black) using Umeyama Sim(3) alignment, with the corresponding ATE reported in the lower-right corner. The selected sequences cover different trajectory lengths, scene scales, and motion patterns. Our method preserves the overall trajectory geometry consistently across the three benchmarks, while the loop-closure backend further reduces accumulated global drift when revisits are detected.

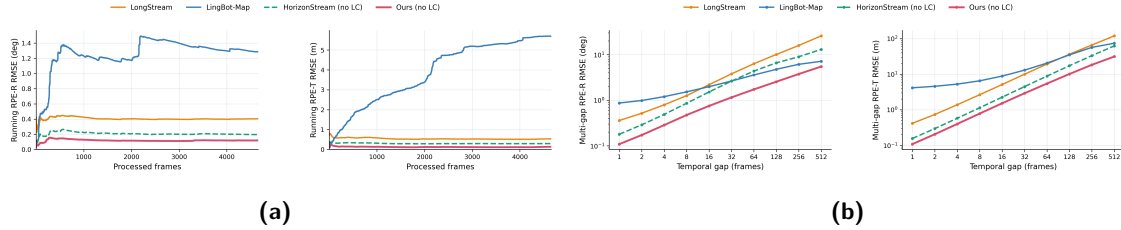


Figure 6. Long-horizon pose stability on KITTI. (a) Running RMSE of adjacent-frame RPE_r and RPE_t on the KITTI-02 sequence as the number of processed frames increases. (b) Multi-gap RPE_r and RPE_t averaged over all KITTI sequences as the temporal gap increases. All methods are evaluated causally without loop closure.

LingBot-Map and HorizonStream. CUT3R and TTT3R attain slightly higher throughput with lower memory usage, but exhibit substantially larger drift over long sequences.

4.3 Dense 3D Reconstruction

Table 7 reports dense reconstruction results on 7Scenes, TUM-Dynamic, and Oxford Spires, covering compact indoor scenes, dynamic RGB-D sequences, and larger landmark-scale environments. Across these different settings, our method achieves competitive reconstruction quality.

On 7Scenes and TUM-Dynamic, our method remains competitive with strong streaming baselines across both CD and F1. Although these sequences can span hundreds or thousands of frames, they are captured within relatively compact indoor environments, where camera observations remain spatially confined and exhibit substantial visual overlap over time. Persistent geometric context can therefore remain relevant throughout the stream. Nevertheless, our method retains comparable reconstruction quality across both static and dynamic indoor scenes without maintaining a persistent global anchor or long-range geometric memory.

On Oxford Spires, the sequences cover substantially larger landmark-scale environments than the compact indoor benchmarks. Our method achieves the lowest CD and the highest F1 among the compared methods. This result indicates that restricting geometric prediction to local temporal context remains effective when reconstruction spans a much larger spatial extent.

Figure 7 further compares the reconstructed geometry on representative sequences of different lengths and scene scales. Across KITTI-02, Oxford Obs.-Q1, and 7Scenes Stairs-01, all three methods recover the main scene structure, while our local formulation maintains coherent geometry from compact indoor scenes to substantially longer driving and landmark-scale sequences. The

Table 6. Streaming efficiency on KITTI-02 (input storage excluded). * marks default multi-frame inference and is not directly comparable to frame-by-frame results.

Method	FPS $\uparrow$	Mem. $\downarrow$
InfiniteVGGT	7.30	15.95
OVGGT	11.87	8.05
Stream3R-w	12.76	5.26
CUT3R	29.62	3.16
TTT3R	25.53	4.65
LongStream	10.36	6.62
LongStream*	23.39	7.50
LingBot-Map	19.74	18.87
HorizonStream	8.02	13.04
HorizonStream*	29.14	26.62
Ours	24.45	6.71

Table 7. Dense reconstruction on 7Scenes, TUM-Dynamic, and Oxford Spires. CD is measured in meters ($\downarrow$), and F1 in percent ($\uparrow$). Best and second-best results are marked in blue and underlined, respectively.

Method	7Scenes ($\tau = 0.25$ m)		TUM-Dyn. ($\tau = 0.25$ m)		Oxford Spires ($\tau = 4$ m)	
	CD $\downarrow$	F1 $\uparrow$	CD $\downarrow$	F1 $\uparrow$	CD $\downarrow$	F1 $\uparrow$
CUT3R	0.18	75.18	0.11	90.68	7.26	41.22
TTT3R	0.11	86.98	0.06	<u>94.45</u>	7.77	40.42
Stream3R-w	0.12	87.57	0.20	87.40	6.44	47.75
InfiniteVGGT	0.08	91.96	0.10	92.94	7.38	47.83
OVGGT	0.07	92.91	0.09	93.91	6.84	55.38
LongStream	<u>0.06</u>	94.60	0.14	88.29	5.69	53.84
LingBot-Map	0.05	<u>96.01</u>	<u>0.08</u>	95.97	<u>1.68</u>	90.58
HorizonStream	0.23	98.22	0.12	90.97	2.00	84.69
Ours	<u>0.06</u>	94.88	0.11	92.19	1.37	91.81

Table 8. Ablation study on KITTI, Oxford, and VBR. The Frames column reports the number of frames in each video clip used for training, while the Model size column reports the number of model parameters. Results are averaged across the evaluation sequences. Lower values indicate better performance, and the best results are highlighted in blue.

Variant	Training configuration				KITTI			Oxford			VBR		
	Frames	Model size	Refiner	$\mathcal{L}_{comp}$	ATE	RPE $_r$	RPE $_t$	ATE	RPE $_r$	RPE $_t$	ATE	RPE $_r$	RPE $_t$
Baseline	32	996M	—	—	56.60	0.19	0.17	10.16	0.19	0.03	52.90	0.83	0.06
+ Comp. loss	32	996M	—	✓	27.66	0.14	0.13	9.40	0.15	0.03	44.72	0.63	0.06
+ Long-seq. training	128	996M	—	✓	22.31	0.11	0.12	5.95	0.13	0.02	36.77	0.62	0.06
+ Rot. Refiner	128	1B	✓	✓	18.25	0.10	0.10	4.35	0.12	0.02	30.14	0.60	0.04

zoomed regions on KITTI and Oxford further show that local geometric structure remains well aligned within the global reconstruction.

4.4 Ablation Studies

To systematically assess the contribution of each model component, we conduct controlled ablation studies on KITTI, Oxford, and VBR, which collectively cover diverse motion dynamics, scene geometries, and sensing conditions. As shown in Table 8, we begin with a baseline trained using only adjacent-frame pose supervision and progressively introduce the composition loss, long-sequence training, and rotation refiner. Following the evaluation protocol described above, we report ATE, RPE $_r$, and RPE $_t$.

Composition-Aware Pose Supervision. Introducing composition-aware supervision yields consistent improvements over the 32-frame baseline across all three datasets. Although the baseline supervises individual adjacent-frame transformations, it does not explicitly constrain the errors that arise when these transformations are recursively composed, leading to long-term trajectory drift. Composition-aware pose supervision mitigates this issue by directly supervising transformations composed across multiple adjacent steps, thereby enforcing relative-pose consistency over different temporal gaps. These results highlight the importance of compositional consistency for limiting long-term trajectory drift, which pairwise pose supervision alone cannot adequately address.

Long-Sequence Training. Increasing the training clip length from 32 to 128 frames further reduces both ATE and relative-pose errors while keeping the local pose-prediction window unchanged. Compared with short-sequence training, ATE decreases by 19.3%, 36.7%, and 17.8% on KITTI, Oxford, and VBR, respectively. These gains suggest that longer training clips provide more extensive and temporally coherent supervision for adjacent-frame motion estimation. They also reduce artificial clip-boundary effects and expose the model to a broader range of local motion patterns, thereby improving the consistency of relative-pose estimates during recursive composition.

Motion-Visual Contextualized Rotation Refiner. The proposed rotation refiner adds only 4.75M parameters, representing approximately 0.48% of the full model, while reducing ATE by 18.2%, 26.9%, and 18.0% on KITTI, Oxford, and VBR, respectively. By jointly leveraging complementary motion and visual cues from recent frames, the refiner predicts more reliable rotation residuals, thereby improving relative-pose consistency during long-horizon trajectory composition.

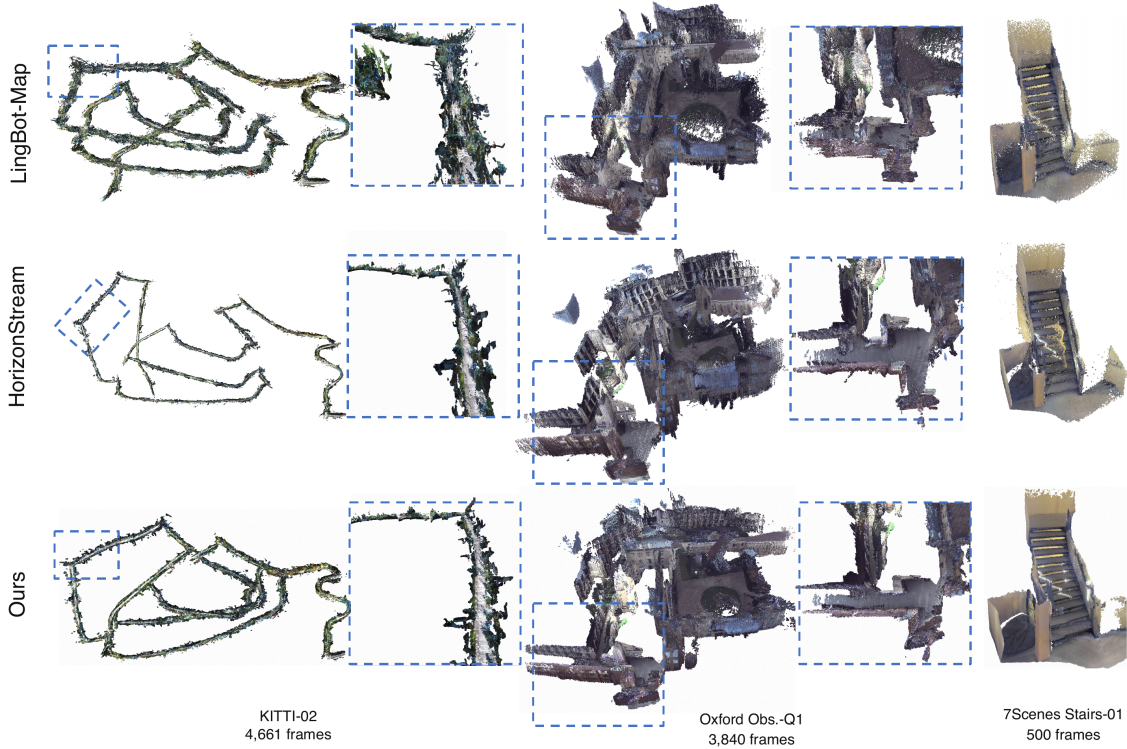


Figure 7. Qualitative dense reconstruction comparison on representative sequences from KITTI, Oxford Spires, and 7Scenes. Dashed boxes indicate regions shown at a larger scale for KITTI-02 and Oxford Obs.-Q1. Across different sequence lengths and scene scales, our method preserves coherent global structure while maintaining well-aligned local geometry using only local temporal context.

5 Discussion

Our results show that the proposed local formulation is particularly effective for long, temporally continuous streams. The model maintains stable local pose estimation as the processing horizon grows and remains consistent under longer pose compositions, leading to strong trajectory accuracy on KITTI, VBR, and Oxford Spires. On compact indoor benchmarks, the gains are less pronounced and our reconstruction results are not uniformly the best. One possible reason is that, in spatially confined scenes with frequent revisits and high visual overlap, persistent geometric context can remain useful over long portions of the sequence. Our method does not explicitly retain such scene-level state, yet remains competitive in these settings, suggesting that the local formulation remains effective across both compact and substantially larger environments.

The optional loop-closure backend complements the local predictor when scene revisits are available. Because it is independent of model training, loop closure introduces sparse long-range constraints only at inference time while leaving the learned streaming formulation unchanged. This provides additional global correction on sequences with useful revisits, without requiring persistent learned long-range state inside the predictor. In this sense, local prediction and loop closure play complementary roles: the former provides a bounded, sequence-length-independent streaming backbone, while the latter offers optional global refinement when additional long-range constraints are available.

The resulting modular design also leaves room for further extensions. The local streaming backbone can incorporate advances in feed-forward reconstruction, caching, and inference acceleration without changing the underlying local-to-global formulation. Future work may explore stronger handling of dynamic scenes [80, 81], external memory [35, 82], larger-scale pretraining [83, 84], and selective long-range constraints for cases where local temporal continuity becomes ambiguous, as well as applications to long-horizon embodied perception [83, 85–88], 3D-VLM [89–91], 3D Generation [92–96] and world modeling [97–99].

6 Conclusion

We presented *ABot-Recon*, a simple streaming 3D reconstruction model whose learned predictor operates entirely within a fixed twelve-frame temporal horizon. Instead of storing or propagating persistent long-range state, *ABot-Recon* predicts geometry in the current camera frame and motion between adjacent frames, keeping the estimation targets local as the sequence grows. Global camera trajectories and scene geometry are recovered by composing these local measurements over time. To make this composition reliable, a lightweight motion-visual rotation refiner improves relative rotations, while composition-aware pose supervision directly constrains their accumulated effect over multiple steps. Experiments on long driving, handheld, and landmark-scale sequences show that *ABot-Recon* delivers accurate camera tracking and dense reconstruction with bounded model memory and per-frame computation. These results demonstrate that persistent learned long-range memory is not a prerequisite for ultra-long streaming reconstruction: a simple short-context model can scale effectively when its local predictions are formulated to remain accurate and reliably composable.

7 Contributions

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A Data Processing

A.1 Temporal Sampling

Video datasets. We construct every training sample as an ordered sequence. For video datasets, the interval $[a, b]$ denotes the range of frame-index steps between consecutive inputs. We use three temporal sampling strategies according to the structure and length of the source trajectory.

1. **Forward fixed-stride sampling.** We uniformly sample one feasible interval from $[a, b]$ and use it throughout the clip. Frames are always traversed in chronological order. This strategy is used for long trajectories that can provide the requested number of frames without reversing direction.
2. **Adaptive forward sampling.** We sample only from the forward portion of a trajectory. When the remaining sequence is too short for the configured maximum interval, we reduce the upper bound to fit the complete clip. Depending on the dataset, the resulting clip uses either one fixed stride or independently sampled per-step strides.
3. **Foldback sampling.** We traverse the source sequence with a stride sampled from $[a, b]$. Upon reaching a sequence boundary, the traversal reverses direction and a new stride is sampled for the next segment. Foldback sampling allows a short source video to provide a long training clip while preserving local continuity and avoiding jumps between unrelated frames.

Multi-view datasets. For unordered multi-view data, we construct fixed-length sequences using a camera pose graph. Let θ_{ij} denote the relative rotation between views i and j , and let $d_{ij} = \|\mathbf{t}_i - \mathbf{t}_j\|_2 / s$ denote their translation normalized by the median nearest-neighbor camera distance s . For each view, we consider its K nearest neighbors and retain at most K_g edges satisfying $\theta_{ij} < \theta_{\max}$ and $d_{ij} < \tau_t$. Beam search then expands valid paths and keeps the best B candidates according to their motion and turning costs. Repeated views are permitted only through valid graph edges, subject to a minimum unique-view ratio $\rho_{\min}$.

Table 9. Pose-graph sequence sampling parameters.

Dataset	$\theta_{\max}$	τ_t	K	K_g	B	$\rho_{\min}$
HyperSim	20°	5	48	24	192	0.50
US4K	20°	5	64	24	128	1.00
BlendedMVS	40°	5	24	14	96	0.75

US4K requires all views to be unique, while HyperSim and BlendedMVS allow constrained revisiting when necessary.

Dataset percentages. The temporal sampling strategy or multi-view construction rule used for each dataset is summarized in Table 10.

A.2 Data Filtering

We find that public datasets contain occasional annotation errors and noisy samples. By inspecting sampled training examples and analyzing the samples associated with loss spikes during training, we identify problematic data and apply dataset-specific filtering. Removing these erroneous samples substantially improves model performance.

BlendedMVS. We observed that some BlendedMVS scenes contain sideways or upside-down images, as shown in Fig. 8. We manually identified and excluded 38 such scenes.



Figure 8. Examples of sideways and upside-down images in BlendedMVS.

TartanAir & TartanAir v2. For the TartanAir datasets, we found that some scenes contained erroneous depth values in sky regions, as shown in Figure 9(a). We invalidated the corresponding depth values using semantic masks. We also observed erroneous water-surface depths in the Ocean scene, as shown in Figure 9(b), and therefore excluded this scene from both TartanAir and TartanAir-v2.

Table 10. Per-dataset sequence-construction and interval policies used for training, resolved from the training configuration and dataset-loader defaults. Interval bounds are measured in source frames. For pose-graph data, b_{med} denotes the scene’s median nearest-neighbor camera baseline.

Dataset	Type	Sampling strategy	Interval / constraint
TartanAir-v2	Synth.	Foldback	Hard [1, 8], Easy [1, 16]
TartanAir	Synth.	Adaptive forward	[1, 20]
OmniWorld-Game	Synth.	Foldback	[4, 16]
TartanGround	Synth.	Foldback	[1, 8]
Virtual KITTI 2	Synth.	Adaptive forward	[1, 5]
HyperSim	Synth.	Pose-graph path	$\Delta R < 20^\circ$, $\Delta t < 5b_{\text{med}}$
PointOdyssey	Synth.	Adaptive forward	[1, 4]
Unreal4K	Synth.	Pose-graph path	$\Delta R < 20^\circ$, $\Delta t < 5b_{\text{med}}$
Spring	Synth.	Foldback	[1, 4]
MVS-Synth	Synth.	Foldback	[1, 4]
Dynamic Replica	Synth.	Adaptive forward	[1, 16]
ASE	Synth.	Foldback	[1, 2]
MidAir	Synth.	Foldback	[1, 8]
MatrixCity	Synth.	Forward fixed-stride	1
AirZoo	Synth.	Foldback	[1, 8]
BlendedMVS	Synth.	Pose-graph path	$\Delta R < 40^\circ$, $\Delta t < 5b_{\text{med}}$
Internal Synthetic	Synth.	Forward fixed-stride	[2, 4]
DL3DV	Real	Adaptive forward	[1, 20]
Waymo	Real	Adaptive forward	[1, 8]
ARKitScenes	Real	Foldback	1
ScanNet++	Real	Foldback	1
WildRGBD	Real	Foldback	[1, 4]
ScanNet	Real	Adaptive forward	[1, 30]
MapFree	Real	Adaptive forward	[1, 30]
ARKitScenes HR	Real	Foldback	1
UASOL	Real	Adaptive forward	[1, 40]
DDAD	Real	Forward fixed-stride	1
KITTI-360	Real	Forward fixed-stride	[2, 6]
Internal Real	Real	Forward fixed-stride	[1, 2]
Holo360D	Real	Foldback	[1, 2]
Holo360	Real	Foldback	[1, 2]

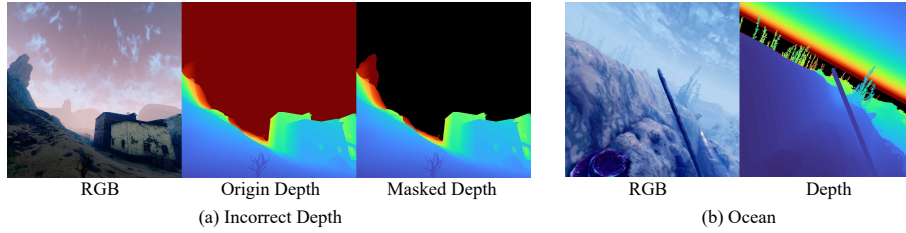


Figure 9. Examples of incorrect scenes in TartanAir and TartanAir-v2.

OmniWorld-Game. We observed erroneous geometric annotations in the dataset, particularly anomalous focal lengths in the camera intrinsics, as shown in Figure 10. We therefore remove unreliable scenes by validating cross-frame correspondences. Specifically, we sample multiple image pairs at different temporal positions in each sequence and obtain reliable 2D correspondences using SIFT [100] feature matching and fundamental-matrix RANSAC. The matched pixels are then back-projected into 3D using the annotated depths and camera intrinsics. We estimate the relative rigid motion between the two frames and compare it with the annotated camera pose. Rotation error, translation direction and scale errors, and depth-normalized geometric residuals are jointly used to identify inconsistent frame pairs. A sequence is removed when severe inconsistencies account for at least 15% of its valid frame pairs and form a cluster of at least three pairs.



Figure 10. Examples of incorrect scenes in OmniWorld-Game.

DL3DV. Since DL3DV is constructed from captured videos using structure-from-motion (SfM) and multi-view stereo (MVS), some scenes contain erroneous pose reconstructions. We assume

that scenes with discontinuous camera trajectories are likely to suffer from reconstruction failures. Therefore, we remove any scene in which the rotation between two consecutive frames exceeds 70° , or the inter-frame translation exceeds $15\times$ median inter-frame translation of that scene. Moreover, because the DL3DV depth maps contain noise, we first discard the farthest 2% of depth values, apply a morphological opening with a 5×5 kernel, and then perform erosion with a 3×3 kernel to remove isolated depth outliers.

HyperSim. We found that all sampled RGB frames from `ai_003_001/cam_00` and `ai_004_009/cam_01` were completely black; therefore, we excluded both sequences from training.

ScanNet++. We observed floating artifacts within the reconstructed meshes of some scenes, which occluded large portions of otherwise valid depth. We therefore excluded the affected scenes from training.

ARKitScenes. We found that the low-resolution depth maps in ARKitScenes, captured using the iPhone LiDAR sensor, are not sufficiently accurate. Therefore, we do not use these depth maps for supervision.

A.3 Spatial Preprocessing and Augmentation

With probability 0.9, we preserve the source field of view by resizing the image isotropically to 504 pixels in width and producing a 504×280 canvas through vertical center cropping or symmetric mean-color padding. The padding value is (0.485, 0.456, 0.406) in normalized RGB space. Geometric annotations and camera intrinsics are transformed consistently with the image. Photometric transformations are applied only to valid image content; synthetic padding is restored to its mean value afterward. The photometric pipeline comprises color jitter (brightness, contrast, saturation, hue, and gamma), JPEG degradation, and image blur. For 20% of training sequences, a single set of augmentation parameters—including the JPEG and blur decisions—is sampled and shared by all frames. For the remaining 80%, these parameters are sampled independently for each frame. This mixed strategy retains temporal coherence in part of the training data while exposing the model to frame-wise photometric variation.

B Runtime and Memory Evaluation Protocol

We evaluate runtime and memory efficiency on the full 4,661-frame KITTI-02 sequence at stride 1 using a single NVIDIA H100 GPU. Each method uses its specified inference precision and input resolution. In addition to frame-by-frame streaming, we report the released chunk-based execution modes of LongStream and HorizonStream, denoted by an asterisk. Before measurement, we perform an untimed 64-frame warm-up and then reset the streaming state. The complete 4,661-frame sequence is subsequently processed for timing. All input frames are preloaded onto the GPU, and image loading and preprocessing are excluded. We synchronize CUDA once immediately before and after the complete forward pass; thus, the reported FPS includes model inference and all streaming-state updates, without introducing per-frame synchronization overhead. Peak memory is reported as the maximum PyTorch-allocated GPU memory excluding the storage occupied by the preloaded input sequence. It includes model parameters, intermediate activations, cached tensors, and persistent streaming state. All methods are evaluated on the same sequence and hardware using the same timing and memory accounting protocol. The resulting throughput and memory measurements are reported in Table 6.

C More Results

We further evaluate the generalization of our method on long videos collected outside the benchmarks in the main paper. These sequences are not associated with accurate ground-truth camera trajectories; therefore, the results in this section are intended as qualitative evidence rather than quantitative comparisons. We consider two complementary settings: low-viewpoint robot navigation and large-scale real-world traversal.

Low-viewpoint robot navigation. Figure 11 presents representative results on videos captured by a quadruped robot. This setting differs noticeably from our training data. The camera is mounted close to the ground, causing the ground plane to occupy a large portion of the image while reducing the visibility of distant scene structures, making frame-to-frame correspondence more challenging than in conventional handheld or vehicle-mounted videos.

Despite this domain shift, our method produces more coherent trajectories across robot-navigation

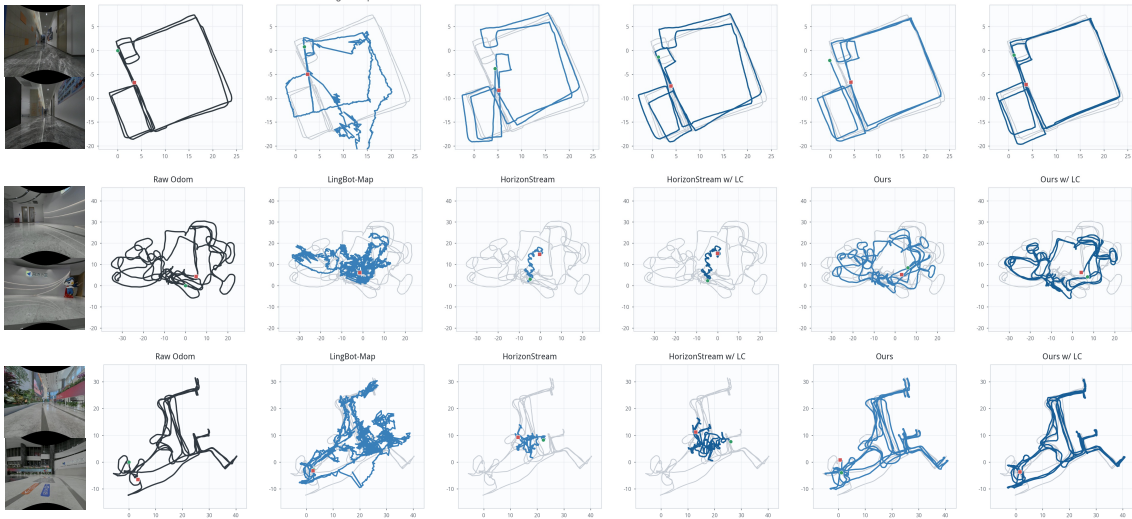


Figure 11. Qualitative camera trajectories on representative quadruped-robot sequences. The low camera height results in ground-dominant observations and motion patterns that differ from the training distribution.

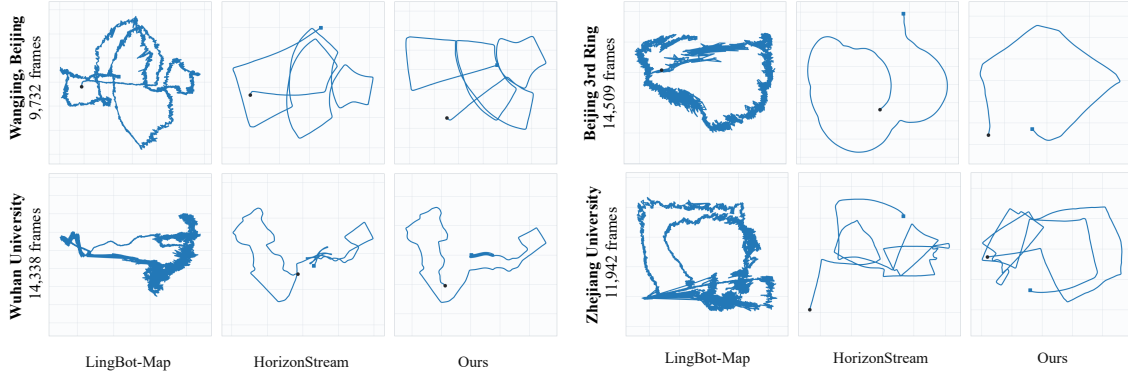


Figure 12. Qualitative no-loop trajectory comparisons on four large-scale real-world videos. For efficiency and fair comparison, all methods are evaluated with temporal stride 5. The results illustrate the behavior of streaming models under variations in scene scale, camera motion, and data distribution.

sequences than other methods. These examples provide qualitative evidence that our method can maintain stable trajectory estimates under camera viewpoints and motion patterns that are less frequently represented in the training data.

Large-scale real-world sequences. We additionally evaluate on four long videos as shown in Figure 12. These sequences cover different scene scales and capture distributions, including extended urban driving, campus roads, repeated structures, intersections, and substantial variations in camera motion and visual appearance. They therefore complement the standard benchmarks with less curated and more heterogeneous long-video streams.

Across the four sequences, our method produces trajectories with more consistent large-scale structure than the compared streaming baselines. This behavior is consistent with the motivation of our design: the network always predicts an adjacent relative transform, so its regression target does not change as the explored area or elapsed sequence length increases. The lightweight temporal correction further suppresses locally correlated rotational errors before they accumulate through pose composition.

Taken together, these qualitative results extend the benchmark evaluation to camera viewpoints, motion patterns, and spatial scales. While accurate ground truth is unavailable for quantitative assessment, the observed trajectory consistency suggests that keeping pose prediction local provides a practical basis for generalization across heterogeneous long-video streams.